

# **INSTRUCTIONS FOR EXAMINER: RUNNING THE SMART INVENTORY EXPERT SYSTEM**

## **1. PREREQUISITES**

Before running the application, please ensure your system meets these requirements:

Operating System: Windows 10 or 11 (the system was developed and tested on Windows)

Python Version: Python 3.10 or higher (tested with Python 3.14.6)

Disk Space: Approximately 200MB free space

Internet Connection: Required for initial package installation

## **2. SET UP INSTRUCTIONS**

### **Step 1: Create Project Directory**

Create a new folder on your desktop named SmartInventory

Open this folder in Windows File Explorer

### **Step 2: Create Required Files**

1. Create requirements.txt:

Right-click in the folder → New → Text Document

Name it requirements.txt (ensure extension is .txt)

Open the file and paste:

`gradio==4.36.0`

`pandas==2.2.2`

`plotly==5.24.1`

`numpy==2.1.0`

`sqlite3`

Save the file

2. Create main.py:

Right-click in the folder → New → Text Document

Name it main.py

Open the file and paste the complete code from the thesis (Section 3.8, Appendix 3)

Save the file

### Step 3: Install Python Dependencies

Open Command Prompt (search for "cmd" in Start menu)

Navigate to your project directory:

```
cd C:\Users\YourUsername\Desktop\SmartInventory
```

3. Install required packages:

```
python -m pip install -r requirements.txt
```

Wait for installation to complete (approximately 2-3 minutes)

### Step 4: Run the Application

1. In the same Command Prompt window, type:

```
python main.py
```

2. Wait for the system to initialize (you'll see:

```
✓ Database initialized with 200 items - All brand mappings corrected!
```

```
🚀 Starting Laitan-Adio Smart Inventory with ALL 200 CORRECTED items...
```

```
* Running on local URL: http://127.0.0.1:7860
```

## 3. ACCESSING THE APPLICATION

Option A (Recommended):

Hold Ctrl and click the URL <http://127.0.0.1:7860> that appears in the terminal

This will open the application in your default web browser

Option B:

Open your web browser (Chrome, Edge, or Firefox)

Paste <http://127.0.0.1:7860> into the address bar

Press Enter

## 4. HOW TO TEST THE SYSTEM

Basic Functionality Test

Go to the "🏠 Browse" tab

Select "Beverages" from the Category dropdown

Select "Carbonated Drinks" from the Subcategory dropdown

Click on any row in the table (e.g., "Coca-Cola Original 50cl")

Verify:

The gauge chart and trend chart appear

The expert advice panel displays detailed procurement analysis

The system correctly identifies stock status

Advanced Functionality Test

Go to the "🔍 Search" tab

Type "Indomie" in the search box

Click on "Indomie Chicken Noodles 70g" in the results

Verify the same charts and expert advice appear

Action Testing

With an item selected, enter a quantity (e.g., 50) in the "Reorder Qty" field

Click "🔄 Reorder"

Verify the confirmation message appears

Click the item again to see updated stock level

## 5. TROUBLESHOOTING COMMON ISSUES

Issue 1: "ModuleNotFoundError"

Symptom: Error message like ModuleNotFoundError: No module named 'gradio'

Solution:

Close the Command Prompt window

Run: `python -m pip install -r requirements.txt` again

Wait for complete installation

Restart the application

Issue 2: "Address already in use"

Symptom: Error message: `OSError: [WinError 10048] Only one usage of each socket address...`

Solution:

Press Ctrl + C in the Command Prompt window to stop the running application

Run python main.py again

Issue 3: Charts not appearing when clicking rows

Symptom: Charts don't display after clicking table rows

Solution:

Close the browser tab

Restart the application with python main.py

Ensure you're using the latest code version with gr.State implementation

Issue 4: Database initialization errors

Symptom: Error related to database initialization

Solution:

Close the Command Prompt window

Delete any .db files in the project folder

Restart the application

## **6. EXAMINER'S VERIFICATION CHECKLIST**

To confirm the system meets all thesis requirements:

Application runs without errors

200 items with correct Nigerian brand mappings are loaded

Browse tab shows hierarchical category/subcategory drill-down

Clicking on items displays gauge charts and trend charts

Chat-style expert advice appears with detailed procurement analysis

Reorder and sale functions work correctly

Search functionality returns relevant results

All 200 items have realistic Nigerian brand mappings

## **7. ADDITIONAL NOTES**

The system is designed to run locally only (no internet required after installation)

No login or authentication is required

The database is automatically created on first run

The application will continue running as long as the Command Prompt window is open

To stop the application, press Ctrl + C in the Command Prompt window

Expected Performance:

Initial load time: 15-20 seconds

Response time to user actions: < 0.1 seconds

System memory usage: ~150-200MB